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5 September 1984
SUPERSEDING
MIL-A-52435C(ME)
19 July 1973

MILITARY SPECIFICATION

AIR CONDITIONER: BASE MOUNTED, AIR-COOLED, 208-VOLT, 3-PHASE, 50- AND 60-HERTZ, AC, SINGLE-PACKAGE, 36,000 BTU/HR

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a self-contained, base mounted, air conditioner. The air conditioners control temperatures and humidities and are suitable for use in extreme hot and humid conditions.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-F-1421	- Fluorocarbon Refrigerants.
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-S-30	- Sacks, Shipping, Paper (Cushioned or Reinforced).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-810	- Enviornmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Material.
MS3108	- Connector, Plug, Electric, Solder Contact, Degree, AN Type.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13219E0790	- Air Conditioner: Base Mounting, Air Cooled, 208 Volt, 3 Phase, 50 and 60 Hertz, AC, Single Package, 36,000 BTU/HR.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN SOCIETY OF HEATING, REFRIGERATING, AND AIR CONDITIONING ENGINEERS (ASHRAE)

Standard 37 - Methods of Testing for Rating Unitary Air Conditioning Equipment.

(Application for copies should be addressed to the American Society of Heating, Refrigrating, and Air Conditioning Engineers, Inc., 1791 Tullie Circle N.E., Atlanta, GA 30329.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, 2501 Northwest Seventh Street, Miami, FL 33125.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 E. Flair Drive, El Monte, CA 91731.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The air conditioner shall be in accordance with Top Assembly TA13219E0790 and as specified herein. The weight of the air conditioner, including a full charge of refrigerant and oil, shall be not more than 530 pounds. The air conditioner shall operate on both 50-hertz and 60-hertz electrical power.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerance is permissable without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the air conditioners. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from the vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.2 First article. Unless otherwise specified (see 6.2) samples shall be subjected to first article inspection (see 4 and 6.3). Any changes or deviations from the first article during production shall be subject to the approval of the contracting officer. Approval of the first article by the activity concerned shall not relieve the contractor of his obligation to furnish air conditioners conforming to this specification (see 6.3)

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. Material not specified in fabrication of the air conditioner shall be compatible to the specified materials and shall be inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the item may be exposed.

3.3.1.1 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make this information available, upon request, to the contracting officer or designated representative.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

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3.3.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the air conditioner may be newly fabricated from recovered materials to the maximum extent practicable, provided the air conditioners produced meet all other requirements of this specification. Used, rebuild or remanufactured components, pieces and parts shall not be incorporated in the air conditioner.

3.4 Performance characteristics.

3.4.1 Capacity. The air conditioner shall have total cooling capacities of not less than the capacities shown in table I when operated at sea level and on 208-volt, 3-phase, 60-hertz electrical power with the evaporator fan operating against a 0.20-inch water gage (WG) static pressure external to the air conditioner.

TABLE I. Cooling capacities.

Con-diton	Condenser Air Inlet Temp Dry Bulb ° F	Evap Air Inlet Temp Dry Bulb °F	Evap Air Inlet Temp Wet Bulb °F	Sensible Cooling Capacity % of Total Cooling Capacity	Minimum Total Cooling Capacity Btu/hr
A	125	90	75	60 to 80	38,300
B	95	80	67	60 to 80	40,700

3.4.2 Operating temperatures. The air conditioners shall be capable of functioning as follows:

a. Start and cycle manually on the cooling cycle at not less than +125° F ambient temperature with air at not less than +125° F, entering the evaporator and condenser.
b. Operate on the cooling cycle without forming frost or ice on the evaporator coil at +60° F ambient temperature, with air at +75° F dry bulb and +63° F wet bulb entering the evaporator.

3.4.3 Evaporator airflow. When the air conditoner is operating on the cooling cycle at the conditions specified in 3.4.1 and table I, the evaporator air flow shall be not less than 875 standard cubic feet per minute (scfm) when the evaporator fan is operating against 0.20-inch WG static pressure external to the air conditioner.

3.4.4 Power consumption. When the air conditioner is operating on the cooling cycle at the conditions specified in 3.4.1 and table I, the total power consumption shall not be more than 8.5 kilowatts (kw) with a power factor of not less than 0.88.

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3.4.5 Operating pressures. When the air conditioner is operating on the cooling cycle conditions specified in 3.4.1 and table I, the condensing pressure shall be not more than 420 pounds per square inch gage (psig) and the suction pressure shall be not less than 60 psig.

3.5 Vibration. When subjected to the test specified in 4.6.2.16 the air conditioner shall show no evidence of structural damage, misalinement or malfunction of components, leaks, fractures, abnormal vibration, or any other irregular operation.

3.6 Storage temperature. The air conditioner shall not be damaged and its operation shall not be impaired by storage at an ambient temperature of +155° F for 4 hours, no solar radiation, and at -65° F for 6 hours, no solar radiation.

3.7 Salt fog and humidity. When subjected to the tests specified in 4.6.2.12 and 4.6.2.13, the air conditioner shall show no evidence of deterioration, corrosion, or change in tolerance limits of any internal or external parts which could in any manner prevent the unit from meeting operational requirements, and shall show no evidence of peeling of paint, malfunctioning, refrigerant leaks, rough operation, damage, electrical shorts, erratic pressure or other irregular operation.

3.8 Rain. When subjected to the tests specified in 4.6.2.11, the air conditioner shall operate without functional damage or impairment of operation. There shall be no penetration of water from the evaporator section to the dry side of the inclosure, or into or on any electrical component (such as motor, electrical connector, terminal switch, or control) that could result in an electrical short, cause damage from rust or corrosion, or become a hazard to personnel.

3.9 Reliability. The specified mean-time-between-failure (MTBF) shall be 960 hours and the minimum acceptable MTBF shall be 480 hours.

3.10 Tilted operation. When subjected to the test specified in 4.6.2.18, the air conditioner shall start and operate on cooling cycle without malfunction, rough operation or vibration, when the base of the air conditioner is positioned 5 degrees from the horizontal in any plane. During tilted operation, all condensate shall be disposed of through the condensate drain line. There shall be no evidence of moisture "carryover" in the airflow leaving the conditioned air outlet.

3.11 Refrigeration system. The refrigeration system shall be of the vapor cycle type, employing refrigerant conforming to BB-F-1421, type 22 and shall be furnished with a full charge in the system. The moisture content in the system shall not exceed 25 parts per million (ppm) by weight in the liquid phase. Refrigerant flow to the evaporator shall be controlled by the solenoid-liquid-shutoff valve and the evaporator thermal expansion valve. When the predetermined temperature is attained within the conditioned space, the thermostat shall close the solenoid-liquid-shutoff valve and stop the flow of refrigerant to the evaporator. The fluid pressure regulator valve in the bypass

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system shall open and bypass the refrigerant gas to the compressor suction and maintain a compressor inlet pressure of not less than 60 psig. An additional expansion valve in the bypass system shall permit a flow of liquid refrigerant to enter the compressor suction line to prevent damage to the compressor motor windings. The entire refrigerant system shall withstand pressures up to 425 psig without permanent deformation of any components. When the discharge side of the refrigerant system is pressurized to 300 psig, any detectable leak shall be not more than 0.1 ounce per year.

3.12 Electrical controls. Electrical controls shall provide automatic operation of the air conditioner. Electrical controls shall consist primarily of a selector switch, a conditioned-air thermostat, and high and low pressure cutout switches. All electrical controls and equipment shall meet the criteria set forth in requirement 1 of MIL-STD-454.

3.12.1 Selector switch. The selector switch shall control operation of the air conditioner as follows. When the switch is in the "OFF" position, all components of the air conditioner with the exception of the compressor crankcase heater shall be de-energized and inoperative. When the switch is in the "VENTILATE" position, the evaporator fan motor shall start and operate. When the switch is in the "COOL" position, the evaporator and condenser fan motors shall start and operate and, after a short time delay, specified in the drawing (see 3.1.1), the compressor motor shall start and operate.

3.12.2 Conditioned-air thermostat. The conditioned-air thermostat shall automatically control the temperature of the conditioned space by regulating the action of the solenoid-liquid-shutoff valve.

3.12.3 High pressure cutout switch. The high pressure cutout switch shall be set to stop the compressor when the condensing pressure reaches 450 psig $\pm$10 psig. The high pressure cutout switch shall be installed to require manual resetting and shall not reset automatically when pressure is decreased.

3.12.4 Low pressure cutout switch. The low pressure cutout switch shall be set to stop the compressor when the suction pressures reduce to 25 psig $\pm$5 psig. The low pressure cutout switch shall be set to automatically start the compressor when the suction pressure increases to 45 psig $\pm$5 psig.

3.13 Lifting attachments. Subjecting each lifting attachment to 2-1/2 times its normal load, imposed with lifting sling leg angles of 45 degrees, shall cause no permanent deformation or breakage to the lifting attachments or to any other part of the air conditioner.

3.14 Wiring. Wiring shall be secured in compact harnesses and attached to frame members with insulated clamps at close intervals to insure a neat and orderly cable run. Cables and wires shall not be spliced at any point throughout the length of their runs. Wiring shall not be attached to panels or other components that require removal during maintenance except for actual electrical connections to components. Electrical isolators shall be used between all wiring and metal components to prevent abrasive action to wire insulation.

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3.15 Electromagnetic interference. The electromagnetic interference emission characteristics of the air conditioner shall not exceed the values given in figures 2 and 3 for steady state conditions. Transient emissions which do not exceed 1 second duration are permitted to exceed the values given in figures 2 and 3 (see 6.9).

3.16 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.16.1, with 1 percent copper 8-quinolinolate or 7 percent salicylanilide (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.16.1 Composition. Composition II shall be used only in the case when local air pollution regulations in the application of varnish precludes the use of compostion I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with rule 442, South Coast Air Quality Management District.

3.17 Treatment and painting. Unless otherwise specified on the drawings, dissimilar metals shall be insulated from one another in accordance with MIL-T-704 and all parts shall be cleaned, treated, and painted in accordance with MIL-T-704, type B with the following exceptions:

a. Internal components previously painted by the manufacturer need not be repainted if they satisfy the salt fog requirements.
b. Coils, refrigerant tubing, electrical components, or other components whose operation will be impaired by painting shall not be painted.
c. The color of the finish coat shall be as specified (see 6.2 and 6.8).

3.18 Accessories. When specified (see 6.2), the following accessories shown on the drawings shall be furnished:

a. Remote control kit assembly.
b. Trash screen.

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c. Left and right side ducts.
d. Air conditioner duct adapters.
e. Duct junction.
f. Air duct hoses.
g. Electrical connector, MS3108-R22-22S.
h. Van duct adapters.

3.19 Maintenance. The availability shall be 95 percent. The mean-time-to-repair (MTTR) for unit maintenance shall be 4 hours, and intermediate shall be 24 hours. The mean-time-between-maintenance (MTBM) for unit maintenance shall be 250 hours, for intermediate shall be 1,000 hours.

3.20 Noise level. The noise level in a conditioned area of a properly engineered installation shall conform to MIL-STD-1474, table II, category E. The condenser side noise level shall conform to MIL-STD-1474, table II, category D.

3.21 Safety. The air conditioner shall have automatic safety controls to protect against overpressure, overheating and electrical overcurrent. The safety devices shall be as specified in 3.7, 3.8, 3.11, 3.12 and 3.14. All reciprocating parts and parts subject to high operational temperatures that are a hazard to operating personnel, shall be guarded or insulated as necessary to eliminate the hazard.

3.22 Human factors. The characteristics of the air conditioner shall conform to section IV of MIL-STD-1472 and shall permit operation and maintenance by personnel who wear clothing appropriate to the environments of interest (see 3.4.2).

3.23 Workmanship.

3.23.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid over-heating. Heated steel shall be allowed to cool slowly. All surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.23.2 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.23.3 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.23.4 Jigs and fixtures. Shop-fabricated components shall be assembled in jigs or frames and joined while held in position. Jigs and frames shall minimize distortion.

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3.23.5 Welders and welding.

3.23.5.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

D1.1 Structural Welding Code, Section 5. Qualifications.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who only make horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, contractor's standard welder qualifications may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welder equipment operators are capable of producing quality welds. The certification shall be made available for review upon request of the contracting officer or the contracting officer's representative.

3.23.5.2 Welding. The surfaces of parts to be welded shall be free from oxide, scale, paint, grease, oil, or any other foreign matter. Immediately before welding, oxide shall be removed by an abrading or a chemical process. Grease and oil shall be removed by wiping with a solvent such as acetone. All surfaces shall be dried after cleaning. Welds shall transmit stress without permanent deformation or failure when the parts connected by welds are subjected to proof and service loadings. Complete and uniform penetration and fusion of the metals shall be obtained on all welds. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding whenever possible.

3.23.6 Brazing. During the brazing operation a small amount of nonoxidizing gas such as dry nitrogen, shall be continuously bled through the refrigerant lines to prevent internal oxide scaling. Care shall be exercised during the brazing operation to protect all components from deformation and damage. The quantity of brazing alloy and flux used shall be no more than that required to make the joint. Flux shall not be present in the refrigerant system and heavy beads and fillets of braze material shall be avoided.

3.23.7 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alternation of the parts themselves or of adjoining parts except for adjustment, and without selection for fit or performance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of

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all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all section 3 and 5 requirements. The inspections in section 4 are the minimum to be used to demonstrate compliance. Quality conformance inspection does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.3 Disassembly inspection. Air conditioners failing a performance test shall be disassembled by the contractor in the presence of a Government representative. Each disassembled part shall be inspected in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, and tolerances. Parts not complying with such requirements shall be rejected.

4.1.4 Parts and components. Parts and components shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings (see 6.5).

4.1.5 End item final inspection requirement (EIFIR). Each air conditioner shall be inspected in accordance with the applicable EIFIR specified on the drawings. One copy of the completed EIFIR shall be shipped with each inspected air conditioner (see 6.6).

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.5).
d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The first article air conditioners shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection and cause for performing the inspection specified in 4.1.3, as applicable.

4.3.2 Tests. One first article air conditioner shall be tested as specified in column 2a of table II and the other first article air conditioner shall be

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tested as specified in column 2b of table II, in the sequence listed, or as otherwise approved by the contracting officer. Failure of any test shall be cause for rejection and for performing the inspection specified in 4.1.3.

TABLE II. Inspection schedule.

Insp Seq	Type of Inspection: First Article		Qual Conf	Insp Comp	Examination or test	Insp Para	Req Para
1	2a	2b	3	4	5	6	7
1	X	X	X	-	Pressure cutout switch tests.	4.6.2.2	3.12.3, 3.12.4
2	X	X	X	-	Refrigerant-system pressure test.	4.6.2.3.1	3.11
3	X	X	X	X	Examination.	4.6.1	3.1, 3.14, 3.16, 3.17, 3.19
4	X	X	X	-	Controls test.	4.6.2.4	3.12
5	X		X1/	-	Refrigerant moisture content test.	4.6.2.5	3.11
6	X		-	-	Electromagnetic interference.	4.6.2.19	3.15
7	X		-	-	Capacity and airflow.	4.6.2.6	3.4
8	X		-	-	Evaporator coil frost.	4.6.2.7	3.4.2
9		X	-	-	High temperature operation.	4.6.2.8	3.4.2
10		X	-	-	High temperature storage.	4.6.2.9	3.6
11		X	-	-	Low temperature storage.	4.6.2.10	3.6
12		X	-	X	Rain.	4.6.2.11	3.8
13		X	-	-	Humidity.	4.6.2.12	3.7
14		X	-	-	Salt fog.	4.6.2.13	3.7
15	X		-	X	Motor protection.	4.6.2.14	DWGS
16		X	-	-	Vibration.	4.6.2.16	3.5
17	X		-	-	Lifting attachment.	4.6.2.15	3.13
18	X		-	-	Tilted operation.	4.6.2.18	3.10
19	X	X	X	-	Production run-in test.	4.6.2.17	3.9
20			-	X	Reliability (MTBF).	4.6.2.20	3.9
21			-	X	Maintenance evaluation.	4.6.2.21	3.19

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TABLE II. Inspection Schedule. (Cont'd)

Insp Seq	Type of Inspection: First Article		Qual Conf	Insp Comp	Examination or test	Insp Para	Req Para
1	2a	2b	3	4	5	6	7
22			-	X	Noise level test.	4.6.2.22	3.20
23			-	X	Safety.	4.6.2.23	3.21
						4.6.2.11	3.8
						4.6.2.3.1	3.11
24			-	X	Human factors	4.6.2.24	3.12
							3.22

1/ Test performed on first production unit. Contractor may certify remaining production units if method of evacuating, dehydrating, and charging of refrigerant system is identical to first production unit.

4.4 Quality conformance inspection.

4.4.1 Examination. Each air conditioner shall be examined as specified in 4.6.2. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each air conditioner shall be subjected to the tests marked "X" in column 3 of table II in the sequence listed. Failure of any test shall be cause for rejection.

4.5 Inspection comparison. The Government may select air conditioners at any time during the contract period and subject these air conditioners to the examination and tests marked "X" in column 4 of table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government on units selected at random from those which have been accepted by the Government and will not include the previously inspected first article air conditioners. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.5.1 Inspection failure. Failure of an inspection comparison air conditioner to meet any requirement specified herein during and as a result of the inspection specified in 4.5 shall be cause for rejection of the inspection comparison air conditioner(s) and shall be cause for refusal by the Government to continue acceptance of production air conditioners until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on air conditioners previously accepted and produced under the contract. Any deficiencies found as a result of the

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inspection comparison will be considered prima facie evidence that all air conditioners accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.6 Inspection procedure.

4.6.1 Examination. The air conditioners shall be examined for the following characteristics:

Number	Characteristic	Requirement Paragraph
101.	Weight not as specified.	3.1
102.	Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for applicable storage and operating environments.	3.3.1
103.	Identification information for materials, material finishes, and treatments are not available as specified.	3.3.1.1
104.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.2
105.	Air conditioner not fully charged with the type of refrigerant as specified.	3.11
106.	Electrical controls not as specified.	3.12
107.	Wiring not installed as specified.	3.14
108.	Electrical circuitry not protected from the effects of moisture and fungus growth.	3.16
109.	Treatment and painting not as specified.	3.17
110.	When required, accessories not as specified.	3.18
111.	Workmanship not as specified.	3.23
112.	Welder certification not provided as specified.	3.23.5.1

4.6.2 Tests.

4.6.2.1 Conditions. Unless otherwise specified herein tests shall be conducted employing 60-hertz electrical power. Unless otherwise specified no braces, sealing materials, tiedown devices, or other items that are not inherently a part of the operating air conditioner shall be employed. Unless otherwise specified herein, no alinement of components, addition of refrigerant, or other servicing shall be performed on the first article model or production air conditioners after testing of the air conditioner has been initiated. Components subject to atmospheric corrosion shall not be treated with any treatment such as grease or oil in order to meet test requirements.

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4.6.2.2 Pressure cutout switches.

4.6.2.2.1 High pressure cutout switch. Prior to installation into the air conditioner, bench test the high pressure cutout switch. Using dry nitrogen, increase the pressure to the high pressure switch until the contacts open. Record the pressure at which cutout occurred. Reduce the pressure to the switch to not more than 300 psig and determine that the contacts remain open. Manually reset the high pressure switch, then determine that the contacts are closed. Repeat the test procedure two more times. Cutout pressures not within the 440 psig to 460 psig range, automatic closing of contacts at reduced pressure, or open contacts after manual reset shall constitute failure of this test.

4.6.2.2.2 Low pressure cutout switch. Prior to installation into the air conditioner, bench test the low pressure cutout switch. Using dry nitrogen, increase the pressure to the low pressure switch until the contacts close. Record the pressure at which cutin occurred. Reduce the pressure until the contacts open. Record the pressure at which cutout occurred. Repeat this test procedure two more times. Cutin pressures not within the 40 psig to 55 psig range or cutout pressures not within the 20 psig to 30 psig range shall constitute failure of this test.

4.6.2.3 Refrigerant system pressure and leakage.

4.6.2.3.1 Refrigerant system pressure. Charge the refrigerant system to 425 psig +0 psig -5 psig with dry nitrogen containing refrigerant gas detectable with a halogen-sensitive electronic leak detector (General Electric Type H-2 or equal). Reduce the pressure to 300 psig ±5 psig. Calibrate the leak detector for the type of gas and percentage of mixture employed and set the detector to indicate a leakage rate of 0.1 ounce per year corrected to the percentage of the mixture. Check the entire refrigerant system for leaks. One or more of the following shall constitute failure of this test:

a. Permanent deformation of any component in the refrigerant system.
b. Any detectable refrigerant leak.

4.6.2.3.2 Refrigerant system leakage. Test for leakage at room ambient temperature (70° F or above) with a halogen-sensitive electronic leak detector (General Electric type H-2 or equal). Calibrate the leak detector for the type of gas and set the detector to indicate a leakage rate of 0.1 ounce per year. Check the entire refrigerant system for leaks. A detectable leak shall constitute failure of this test.

4.6.2.4 Controls. Conduct the controls test at an ambient temperature between +65° F and 75° F. Position the selector switch in the "OFF" position. Determine that the air conditioner components are de-energized and inoperative, with the exception of the compressor heater. Adjust the thermostat to the maximum cool setting and position the selector switch in the "FAN" position. Determine that the evaporator fan motor is energized and operating, and that the

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condenser fan motor and compressor motor are de-engergized. With the thermostat at the maximum cool setting, position the selector switch in the "COOL" position. Determine that the evaporator and condenser fan motors are energized and operating and, after a time delay, that the compressor motor is energized and operating. Determine the length of the time delay. Determine that there is a decrease in temperature of the evaporator-air-discharge. With the selector switch in the "COOL" position, adjust the thermostat to the maximum warm setting. Determine that the compressor and the fan motors continue operation. Determine that the liquid line solenoid valve closes to stop flow to the expansion valve, that the suction pressure regulating valve opens to maintain the suction pressure at not less than 60 psig, and that the by-pass line expansion (quench) valve permits flow to the compressor suction line. Determine that there is an increase in the temperature of the evaporator-air-discharge. Nonconformance to 3.12 shall constitute failure of this test.

4.6.2.5 Refrigerant moisture content. After not less than 1 hour of continuous operation in the cooling mode in an ambient temperature of not less than +95° F, shut down the air conditioner. After an additional period of 1 hour, withdraw a sample of refrigerant vapor from the low-pressure side and determine the moisture content by one of the methods specified in BB-F-1421. A moisture content of more than 25 ppm shall constitute failure of this test.

4.6.2.6 Capacity and airflow. The capacity and airflow test shall be conducted utilizing the instruments, apparatus and procedures specified in ASHRAE standard 37. The test methods from ASHRAE standard 37 shall be the Air-Enthalpy Method for both the indoor side and the outdoor side. The air conditioner shall be installed in an outside room with the evaporator air discharge and return air inlets ducted to an inside room. Test the air conditioner for cooling capacity at the capacity rating conditions specified in 3.4.1. Insulating or sealing material that is not inherently a part of the air conditioner shall not be employed except that ducts and duct adapters may be insulated. Nonconformance to 3.4.1, 3.4.3, 3.4.4, or 3.4.5 shall constitute failure of this test.

4.6.2.7 Evaporator coil frost. Operate the air conditioner continuously for 12 hours in the cooling mode in an ambient temperature of +60° F with air at +75° F dry bulb and 63° F wet bulb entering the evaporator. Immediately following the 12-hour operation, inspect the evaporator coil. Any evidence of ice or frost on the evaporator coil, or failure of the air conditioner to operate and cool continuously for 12 hours, shall constitute failure of this test.

4.6.2.8 High temperature operation. Subject the air conditioner to a temperature of +125° F +3° F -0° F for a period of 4 hours. After completion of the exposure period, operate the air conditioner in the cooling mode for a period of not less than 1 hour in an ambient temperature of +125° F +3° F -0° F with air at +125° F +3° F -0° F entering the evaporator and condenser. Turn the air conditioner off and restart and operate the air conditioner for a period of not less than 15 minutes. Repeat the "OFF" and 15 minute-operating-period cycle

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two more times. Repeat the above test operating on 50 hertz electrical power. Inability of the air conditioner to start, operate, restart, or nonconformance to 3.4.2 shall constitute failure of this test.

4.6.2.9 High temperature storage. Subject the air conditioner to a temperature of +155° F for a period of 4 hours. At the conclusion of the exposure period remove the air conditioner from the test chamber and examine for deterioration, corrosion or damage. After completion of the examination, allow the air conditioner to soak at room ambient temperature for a period of not less than 1 hour but not more than 3 hours. After completion of the soaking period, operate the air conditioner on cooling cycle for a period of not less than 1 hour. Inability of the air conditioner to start or operate, or any evidence of deterioration, corrosion, damage, such as rapture of tubing, damage to control components, binding of motors, fans, or other mechanical parts, deterioration of gaskets, damage to insulation, breakdown of adhesives, permanent setting of gaskets or packing, or loss of refrigerant shall constitute failure of this test.

4.6.2.10 Low temperature storage. Subject the air conditioner to a temperature of -65° F for a continuous period of 12 hours. At the conclusion of the exposure period, remove the air conditioner from the test chamber and examine for deterioration and damage. After completion of the examination connect power to the air conditioner and allow the air conditioner to soak at room ambient temperature for a period of not less than 6 hours. After completion of the soaking period, operate the air conditioner on the cool mode for a period of not less than 1 hour. Any evidence of deterioration or damage, or failure of the air conditioner to start and operate shall constitute failure of this test.

4.6.2.11 Rain. The air conditioner evaporator section shall be either flush with or protrude through the opening of an inclosure. Not more than 4 inches of the air conditioner shall protrude through the opening. Seal the opening to prevent any water from entering the inclosure. Connect a suitable external drain line to the condensate drain connection of the air conditioner. Rainfall shall be produced by means of water spray and shall be measured with U.S. Weather Bureau type gages. Nozzles shall emit droplets having a diameter of not less than 1.5 millimeters. Water supply temperature shall be between 50° F and 70° F and shall contain flourescent dye for ease in determining the locations and extent of leakage. After each rain test, remove all panels, access doors, and component covers from the air conditioners as required to examine for water penetration. After the examination, operate the air conditioner for 15 minutes in the cooling mode.

4.6.2.11.1 Free rain. With the air conditioner not operating, subject each of the three exposed sides and top of the air conditioner to a simulated rainfall of 4 inches per hour ±1 inch for 30 minutes each, for a total test duration of not less than 2 hours. Direction of rainfall against each of the three sides shall be varied throughout the test from vertical to 45 degrees from vertical.

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4.6.2.11.2 Blowing rain. With the air conditioner operating on cooling mode, subject each of the three exposed sides to a simulated rainfall of 1 inch per hour ±1/2 inch combined with a horizontal wind of 40 miles per hour ±5 miles per hour for a period of 1 hour. Total test duration shall be not less than 3 hours.

4.6.2.11.3 Basis of failure. One or more of the following shall constitute failure of the rain test:

a. Evidence of water leaking from the evaporator section to the dry side of the inclosure.
b. Malfunctioning of components or an electrical short during operation.
c. Nonconformance to 3.8.

4.6.2.12 Humidity. Subject the air conditioner to humidity in accordance with MIL-STD-810, method 507.2, procedure III. The air conditioner shall be operated in accordance with 4.6.2.4 prior to and after the exposure period to determine satisfactory functioning of controls and components. The front side of the control box shall be covered and sealed during exposure. Do not operate the air conditioner during exposure and do not expose the air conditioner to extreme temperature prior to the test. After exposure, the junction box may be removed and the magnetic iron in the relays may be cleaned with a wire brush to remove rust and scale from the laminated surfaces. Nonconformance to 3.7 shall constitute failure of this test.

4.6.2.13 Salt fog. Subject the air conditioner to a salt fog environment in accordance with MIL-STD-810, method 509.2, procedure I. The front side of the control box shall be covered and sealed during exposure. Do no operate the air conditioner during exposure. Prior to this test and after the exposure period, operate the air conditioner in accordance with 4.6.2.4 to determine satisfactory functioning of the controls and components. After exposure, the junction box may be removed and the magnetic iron in the relays may be cleaned with a wire brush to remove rust and scale from the laminated surfaces. Nonconformance to 3.7 shall constitute failure of this test.

4.6.2.14 Motor winding thermal overload protectors.

4.6.2.14.1 Measurements. Insulation resistance, motor winding resistance, motor winding temperature and motor electric current shall be measured for this test. Insulation resistance between the motor windings and frame of the motor will be measured with a megger insulation tester. Motor electric current will be measured with a "clamp on" type ammeter. Motor winding temperature shall be calculated from the winding resistance, as follows:

$$t = \frac{(234.5 + t_1)\, R_2 - 234.5 R_1}{R_1}$$

where: t_1 = ambient temperature - degrees celsius.
R_1 = resistance of windings at t_1 temperature - ohms.
R_2 = resistance of windings at the time the motor winding temperature is to be determined - ohms.
t = the motor winding temperature to be determined - degrees celsius.

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Motor winding resistance shall be measured by the Seely Circuit method with the motor energized or by the cooling curve method with a wheatstone bridge after the motor has been de-energized. For tests using the cooling curve method, the motor shall be connected to its power source through a three-pole, double throw (3PDT) switch to enable isolation of the windings from the energized circuit. When measuring by the cooling curve method, isolate the energizing circuit from the motor with the 3PDT switch and immediately measure and record the resistance of each winding phase. Repeat the resistance measurement/recording at 15 second intervals for a period of five minutes. With the data obtained, plot a curve of resistance versus time, then by extrapolation determine the resistance value at time zero. Use the resistance at time zero to calculate the winding temperature.

4.6.2.14.2 Compressor motor. Place the air conditioner in an ambient temperature of 70° F ±1° F for 24 hours. Measure and record the ambient temperature and the resistance of each phase winding of the motor. Place jumper wires around the circuit breaker to prevent de-energizing the power supply and disconnect power to the condenser fan motor. Immerse the liquid quenching expansion valve bulb in water at a temperature of not more than 35° F. Position the mode selector switch in the "COOL" position and adjust the conditioned air thermostat at the maximum warm setting. Operate the air conditioner until the thermal overload protection device for the compressor motor windings is activated. Measure the resistance of the compressor motor windings. Calculate the temperature of the windings as specified in 4.6.2.14.1. Check electrical continuity from phase to phase. Remove the jumper wires from the circuit breaker, reconnect power to the condenser fan motor, re-install the liquid quenching expansion valve bulb, and operate the air conditioner in the cooling mode with the thermostat adjusted to the maximum cool setting for 1 hour in an ambient temperature range of 80° F to 90° F. One or more of the following shall constitute failure of this test:

a. Motor winding temperature (t) in excess of motor manufacturer's recommended limit.
b. Lack of electrical continuity from phase to phase.
c. Inability of the air conditioner to operate without malfunction.

4.6.2.14.3 Evaporator fan motor. Lock the fan motor shaft to prevent rotation. Measure the insulation resistance, motor winding resistance, and motor winding temperature. The motor winding temperature shall be 70° F ±5° F for not less than 24 hours before beginning the test. Place the selector switch to "FAN" position, energizing the fan motor with the shaft locked. Continue operation with the fan motor energized until the temperature current overload protective device has cut-out and cut-in three times. Measure and record the locked-rotor current and motor winding resistance. Determine the motor winding temperature in accordance with 4.6.2.14.1. At the completion of the test, de-energize the motor and allow the motor winding temperature to equalize at a temperature of 70° F ±5° F for not less than 24 hours. Measure the motor winding resistance and insulation resistance. Check the electrical continuity through each phase winding. Remove the locking device from the evaporator fan motor shaft. Operate the air conditioner in the "FAN" mode for 30 minutes.

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4.6.2.14.4 Condenser fan motor. Lock the fan motor shaft to prevent rotation. Measure the insulation resistance, motor winding resistance, and motor winding temperature. The motor winding temperature shall be 70° F ±5° F for not less than 24 hours before beginning the test. Electrically disconnect the compressor motor and connect a jumper wire across the motor thermostat terminals of the connector. Place the selector switch to "COOL" position, energizing the condenser fan motor with the shaft locked (the evaporator fan will operate normally). Continue this operation of the air conditioner until the temperature current overload protective device has cut-in and cut-out three times. Measure and record the locked-rotor current and motor winding resistance. Calculate the motor winding temperature as specified in 4.6.2.14.1. At the completion of the test, de-energize the motor and allow the motor winding temperature to equalize at a temperature of 70° F ±5° F for not less than 24 hours. Measure the motor winding resistance and insulation resistance. Check the electrical continuity through each phase winding. Remove the locking device from the fan motor shaft, remove the jumper wire, reconnect the compressor motor and operate the air conditioner in the cooling mode for 30 minutes.

4.6.2.14.5 Basis of failure for the fan motors. One or more of the following shall constitute failure of the fan motor thermal overload protection test:

a. A motor winding resistance at 70° F varying more than 15 percent from the initial motor winding resistance.
b. An insulation resistance of less than 12 megohms.
c. Inability of the air conditioner to operate without any malfunction.
d. A motor winding temperature in excess of motor manufacturer's recommended limit.
e. Lack of electrical continuity through any phase winding.
f. Any evidence of smoke (visual) emanating from the motor.
g. Motor running current more than 20 percent above nameplate amperage.
h. Locked rotor current varying by more than 25 percent from the manufacturer's furnished data.

4.6.2.15 Lifting attachments. Lift and suspend the air conditioner with a sling connected to the lifting attachments. The legs of the sling shall be maintained at an angle of 45 degrees +5 degrees -0 degree from the vertical. Measure the load imposed on each lifting attachment. Lower the air conditioner to the floor or ground and subject each lifting attachment to a pull of not less than 2-1/2 times the measured load inposed on each attachment along the sling angle specified herein. Permanent deformation, rupture, cracking or breakage of any of the lifting attachments or any other part of the air conditioner shall constitute failure of this test.

4.6.2.16 Vibration. Subject the air conditioner to sinusoidal vibration applied along each of the three mutually perpendicular axes. The vibration acceleration levels and double amplitudes specified in figure 4 shall be maintained at the air conditioner mounting base. The vibration shall be logarithmically cyled from 5 cycles per second (cps) to 200 cps and return to 5 cps over a 12 minute sweep period. The ascending sweep time and the descending

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sweep time shall be equal (6 minutes each). The total test time per axis shall be 60 minutes. The air conditioner shall not be operated during the vibration test. The air conditioner shall be mounted by bolts utilizing the mounting receptacles provided in base of the air conditioner. Auxiliary braces, straps, or other tiedown devices, other than those specified herein, shall not be employed to secure the air conditioners to the test apparatus. No blocking or tiedown of internal components other than those that are a part of the air conditioner shall be employed . Removable panels and access doors shall be in place during all testing. With the air conditioner mounted, there shall be no resonance inherent in the test apparatus at the frequencies and acceleration levels specified for this test. During the tests, the base of the air conditioner shall remain in a horizontal plane. Mount accelerometers on the test apparatus immediately adjacent to the four corners of the base of the air conditioner. The four accelerometers shall be connected into an electric or electronic network which will average the accelerometers output signals into a single indicated g level of the magnitude specified for this test. After completion of each test axis, conduct the control test specified in 4.6.2.4 and the refrigeration system leak test specified in 4.6.2.3.2. Nonconformance to 3.5, 4.6.2.3.2, or 4.6.2.4 shall constitute failure of this test.

4.6.2.17 Production run-in.

4.6.2.17.1 Production performance. Production performance tests shall be conducted as specified herein utilizing instruments specified in ASRAE standard 37. Install ducts on the evaporator return air openings, evaporator air discharge openings, and condenser air inlet opening of each air conditioner. Each duct shall be not less than 12 inches long and of the same cross-sectional dimensions as the respective openings. Equip each duct with a thermocouple grid to measure average entering and leaving air temperatures to the evaporator and entering air to the condenser. Measure the evaporator air wet bulb temperature by a psychrometer device located in the test chamber. Operate the air conditioners for not less than 1 hour in the cooling mode while maintaining the supply voltage to the air conditioner within ±5 percent. The air temperature entering the evaporator return and condenser inlet shall be as specified in table V. After stabilization, three sets of the data specified in table IV shall be recorded and averaged during the last 30 minutes of operation (every 10 minutes). After completion of the above, operate the air conditioner in an ambient temperature between 100° and 120° F for a total period of 48 hours in the cooling mode, in cyclic operation of 55 minutes "ON" and 5 minutes "OFF". After completion of the 48 hours operation, continue operating the air conditioner in the cooling mode for not less than an additional hour while maintaining the supply voltage to the air conditioner within ±5 percent and temperature of the air entering the evaporator return and condenser inlet as specified in table V. After stabilization, three sets of the data specified in table IV shall be recorded and averaged during the last 30 minutes of operation (every 10 minutes). After completion of the 30 minute operation period, conduct the refrigeration system leak test as specified in 4.6.2.3.2.

4.6.2.17.2 Production performance curves. Not less than 25 units shall be used to establish acceptance curves. When less than 25 units are produced all production units will be used to establish acceptance curves. Data obtained in

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accordance with 4.6.2.17.1 shall be used by the contractor to select one high and one low unit which shall then be capacity tested in accordance with 4.6.2.6 at condition "A" as shown in table I. The contractor shall select the high and low units based on evaporator temperature change with additional consideration given to suction and discharge pressures and kilowatt power input. If any high or low limit air conditioner fails to meet the requirements of 3.4.1, 3.4.4 or 3.4.5 it shall be replaced by the next low or high unit and that unit tested. Operate the high and low air conditioners continuously in the cooling mode for not less than 1 hour under each of the conditions specified in table III. After stabilization, three sets of data specified in table IV shall be recorded and averaged for each of the conditions in table III during the last 30 minutes (every 10 minutes). Establish high and low units of acceptance for production units by plotting the following curves from data required by table IV:

a. Average evaporator entering dry bulb temperature vs. average temperature differential between evaporator inlet and outlet air temperatures for each of the condenser entering temperatures recorded.
b. Average evaporator entering dry bulb temperature vs. kilowatt power input for each of the condenser entering temperatures.
c. Average evaporator entering dry bulb temperature vs. suction pressure for each of the condenser entering temperatures.
d. Average evaporator entering dry bulb temperature vs. discharge pressure for each of the condenser entering temperatures.

TABLE III. Temperature conditions for production performance curves.

Location	Temperature			
	Condition I	Condition II	Condition III	Condition IV
Evaporator return	87° F ±.5° F	93° F ±.5° F	87° F ±.5° F	93° F ±.5°F
Condenser inlet	107° F ±.5° F	113° F ±.5° F	113° F ±.5° F	107° F ±.5°F
Evaporator return and condenser inlet: Dry coil condition, all conditions.				

TABLE IV. Data required for production performance curves.

a. Dry bulb temperature of air entering the evaporator, °F.
b. Wet bulb temperature of air entering the evaporator, °F.
c. Dry bulb temperature of air leaving the evaporator, °F.
d. Dry bulb temperature of air entering the condenser, °F.
e. Condensing pressure, psig.
f. Suction pressure, psig.
g. Power consumption of the air conditioner, watts.

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TABLE V. Temperature conditions for production performance.

Location	Temperature
a. Evaporator return	Dry bulb - 90° F ±3° F
b. Evaporator return	Wet bulb - dry coil condition
c. Condenser inlet	Dry bulb - 110° F ±3° F
d. Condenser inlet	Wet bulb - dry coil condition

4.6.2.17.3 Failure criteria. One or more of the following shall constitute failure of this test:

a. Temperature differential between the average air dry bulb temperature entering the evaporator return and the average air dry bulb temperature leaving the evaporator recorded before and after the 48 hour test phase in 4.6.2.17.1 not within the curves developed in 4.6.2.17.2.
b. Average condensing pressure recorded before and after the 48 hour test phase in 4.6.2.17.1 not within the curves developed in 4.6.2.17.2.
c. Average suction pressure recorded before and after the 48 hour test phase in 4.6.2.17.1 not within the curves developed in 4.6.2.17.2.
d. Average power consumption on cooling recorded before and after the 48 hour test phase in 4.6.2.17.1 not within the curves developed in 4.6.2.17.2.
e. Failure of any component, or fracture or breakdown of any material.
f. Evidence of refrigerant leak greater than 0.1 ounce per year.
g. Need for any servicing of the air conditioner during operation.

4.6.2.18 Tilted operation. Operate the air conditioner in the cooling mode for a period of not less than 12 hours in an ambient temperature of +80° F ±3° F dry bulb and +75° F ±3° F wet bulb with air at the same temperature entering the evaporator. During this test, operate the air conditioner with the bottom at a slope of 5 degrees from the horizontal. For each 3 hours of operation, the bottom edge of the four sides shall be raised, in turn, to form the 5-degree slope. Nonconformance to 3.10 shall constitute failure of this test.

4.6.2.19 Electromagnetic interference. The air conditioner shall be tested in accordance with test methods RE-02 (0.15 MHz to 1000 MHz) and CE-03 (1.5 MHz to 65 MHz) of MIL-STD-461 as follows:

a. Steady state conditions. Controls of the air conditioner shall be set for the following modes of opertion at steady state conditions:

Mode (selector switch) setting	Thermostat setting
Cool	Cooler (lowest temperature)
Cool	Warmer (highest temperature)

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b. Transient conditions. Transient durations shall be measured when the air conditioner controls are switched between the following modes:

Mode	Selector switch setting	Thermostat position
Cool to bypass	Cool	Cooler to warmer (low to high temperature)
Bypass to cool	Cool	Warmer to cooler (high to low temperature).

4.6.2.19.1 Basis of failure. One or more of the following shall constitute failure of the electromagnetic interference test:

a. Steady state emissions in excess of figures 2 and 3.
b. Transient emissions exceed the limits specified in figures 2 and 3 and exceed 1 second duration.

4.6.2.20 Reliability. Using the MTBF specified in 3.9, the air conditioners shall be tested as specified herein with "accept" and "reject" criteria in accordance with MIL-STD-781, test plan IV. The air conditioners shall be considered to have passed this test if the test times and failures fall within the "accept" region and considered to have failed if they fall within the "reject" region of test plan IV, and testing shall be terminated. If the assessment falls within the "continue test" region, multiples of the previous testing shall be performed until an "accept" or "reject" assessment is achieved.

4.6.2.20.1 Test facility. Install the air conditioners in a test chamber consisting of two rooms. One room shall function as an indoor room and shall maintain the evaporator return air temperature as specified in figure 1 within ±5° F. The second room shall function as an outdoor room and shall maintain the condenser return air temperature as specified in figure 1 within ±5° F. These temperatures shall be maintained on a 24 hour cyclic basis as shown in figure 1 throughout the reliability test.

4.6.2.20.2 Operation. All air conditioner controls such as selector switch and thermostat shall be operated manually or automatically. If automatic means are used, they shall not interrupt the control circuit but shall physically operate the air conditioner control components. The controls shall be positioned as shown in table VI.

TABLE VI. Control positions for reliability test.

Mode	Time	Control	Control Position
Ventilate	0030 to 0130	Selector Switch	Ventilate
Bypass	0130 to 0400	Selector Switch	Cool
Bypass	0130 to 0400	Thermostat	Maximum warm
Cooling	0400 to 2100	Selector Switch	Cool
Cooling	0400 to 2100	Thermostat	Maximum cool
Bypass	2100 to 2330	Selector Switch	Cool
Bypass	2100 to 2330	Thermostat	Maximum warm
Off	2330 to 0030	Selector Switch	Off

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4.6.2.20.3 Data. The following shall be monitored and recorded throughout the test as required to determine that normal air conditioner operation is being maintained and to identify problems in event of failure or malfunction:

a. Voltage.
b. Current.
c. Frequency.
d. Refrigerant pressures.

    1. Suction.
    2. Discharge.

e. Refrigerant temperatures.

    1. Compressor discharge.
    2. Liquid line.
    3. Suction line (at compressor).

f. Air temperatures.

    1. Evaporator return (db).
    2. Evaporator discharge (db).
    3. Condenser return (db).

g. Abnormal operation (visual and audio observation).
h. Sight glass observation (clear or bubbles).
i. Identification of unit.
j. Initials of technician.
k. All failures.
l. Time of day of failure.
m. Action taken if failure occurs.

4.6.2.20.4 Basis of failure. A failure shall be defined as any malfunction which cannot be remedied by adjustment, repair, or replacement action within 1 hour using the air conditioner controls, Tool Kit, Air Conditioner, SC5180-97-CL-E49, and Tool Kit, Service, Refrigeration Unit, SC5180-97-CL-E18.

4.6.2.21 Maintenance evaluation. The maintenance repair times and the mean times between maintenance shall be determined during reliability testing (see 4.6.2.20). All maintenance actions shall be assessed during the reliability test and other inspection comparison testing. Nonconformance to 3.19 shall constitute failure of this test.

4.6.2.22 Noise level. The air conditioner shall be mounted in the wall of a shelter or building with the front face of the unit (evaporator outlet grille side) flush with the inside surface. Ducts shall be installed to connect the inside room to the air conditioner return air openings. The air conditioner shall be operated in the cooling mode and sound data as required by MIL-STD-1474 shall be measured, recorded and reported for three positions in the inside room and three positions outside of the shelter or building. The data shall include

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dB(A), dB(C) and octave band pressure levels for background and test conditions. The three inside positions shall be at an elevation equal to the centerline of the evaporator discharge grilles and at a distance of 48 inches from the front face of the air conditioner. One position shall be on the centerline of the air conditioner, perpendicular to the front face and the other two indoor positions shall be at a 30 degree angle to the left and to the right of the perpendicular centerline. The three positions for the outside shall be referenced at the mid-point of the rear face (condenser inlet opening) of the air conditioner. One position shall be on the centerline perpendicular at 48 inches from the rear face and the other two positions shall be at 45 degree angles from the centerline. Nonconformance to 3.20 shall constitute failure of this test.

4.6.2.23 Safety evaluation. Evaluation shall be accomplished throughout testing. Nonconformance to 3.21 shall constitute failure of this test.

4.6.2.24 Human factors. The air conditioner shall be evaluated to determine that the human factors criteria have been met. Nonconformance to 3.22 shall constitute failure of this test.

4.7 Inspection of packaging.

4.7.1 Examination of first article pack. The first article pack shall be examined for the defects specified in 4.7.2.3. Presence of one or more defects shall be cause for rejection.

4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects. The AQL shall be 2.5 percent defective.

113.	Preservation methods not as specified for level A or level B.
114.	Air conditioner not secured to base of box as specified for level A and level B.
115	Technical publications not secured to top of air conditioner for level A or level B.
116.	Remote control kit assembly, ducts, duct adapters, and air duct hose, when included, not packed together for level A and level B.
117.	Marking illegible, improper, or incorrect for level A, level B or commercial.

5. PACKAGING

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5.1 First article pack. Unless otherwise specified (see 6.2) the contractor shall furnish a first article pack for examination within the time frame specified, to prove prior to starting production that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The first article pack may be accomplished utilizing either the first article air conditioner or production air conditioner. If the first article air conditioner is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article air conditioner and the production air conditioner.

5.2 Preservation. Preservation shall be level A, level B, or commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Air conditioner. The air conditioner shall be preserved in accordance with MIL-P-116, method IIa. The air conditioner shall be secured to the bottom of the box specified in 5.3.1 by bolting directly through the barrier material into the mounting nuts provided in the frame of the air conditioner.

5.2.1.2 Technical publications. Technical publications shall be inclosed in a polyethylene envelope which shall be closed by heat sealing. The envelope shall be secured to the top of the air conditioner with tape prior to enclosing the unit in the method II barrier.

5.2.1.3 Remote control kit assembly. The remote control kit assembly, when provided, shall be preserved in accordance with MIL-P-116, method IIe.

5.2.1.4 Electrical connector. The electrical connector shall be packaged in a sack conforming to PPP-S-30, type II, size as appropriate. The sack shall be stapled shut.

5.2.1.5 Ducts and duct adapters. Ducts and duct adapters, when provided, shall be nested together, whenever possible, and secured with tape or other suitable means.

5.2.2 Level B.

5.2.2.1 General. Technical publications and accessory items (remote control kit, electrical connector, ducts and duct adapters) shall be preserved as specified for level A; technical publications shall be secured to the air conditioner as specified in 5.2.1.2.

5.2.2.2 Air conditioners. Air conditioners shall be covered with a polyethylene film shroud of not less than 3 mils in thickness. The shroud shall be secured to the air conditioner with tape.

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5.2.3 Commercial. Each complete air conditioner shall be preserved in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A, level B or commercial as specified (see 6.2).

5.3.1 Level A. The air conditioner, preserved as specified in 5.2.1.1, shall be packed in a box conforming to PPP-B-601, overseas type. The box shall be modified by the addition of two 3- by 4-inch skids placed so that the bolts securing the air conditioner to the box base pass through the skids. The boltheads shall be countersunk into the skids. The inside dimensions of the box shall be such that a 1-inch clearance is provided between the method II barrier and the inside surfaces of the box. The top of the box shall be secured with wood screws instead of nails. The box shall be reinforced with strapping conforming to QQ-S-781, class 1, type I or IV, size as applicable; and unless otherwise specified shall be finish B (see 6.2). When specified finish A strapping shall be used.

5.3.1.1 Accessories. When provided, the remote control kit assembly, electrical connector ducts, duct adapters, junction, and air duct hose, preserved as specified in 5.2.1 or 5.2.2, shall be packed together in a snug-fitting box conforming to PPP-B-601, overseas type. Air duct hose shall be compressed to the maximum extent possible. All items shall be secured within the box with fiberboard pads or cells or cushioning material to prevent movement. The box shall be closed and reinforced in accordance with the appendix to the box specification; strapping shall be as specified in 5.3.1.

5.3.2 Level B. The air conditioner preserved as specified in 5.2.2 shall be bolted to a 5/8-inch-thick plywood base and fitted with a cover conforming to PPP-B-640, class 2, style E, without bottom flaps. The length and width of the base shall be 2 inches greater than the length and width of the air conditioner. Two 3- by 4-inch skids shall be placed on the bottom side of the base; skids shall be located so that the bolts securing the air conditioner to the base pass through them. The boltheads shall be countersunk into the skids. Corner pads 1/2-inch thick with 4-inch-wide legs shall be secured to all vertical edges and to all horizontal top edges of the air conditioner. Wood cleats 1/2-inch wide by 1-inch thick shall be secured with nails to the top surface of the base directly against the air conditioner and between the vertical edge corner pads. The fiberboard cover shall fit snugly over the vertical edge corner pads and rest upon the top surface of the base. The cover shall be stapled to the wood cleats secured to the top surface of the base. The pack shall be with straps as specified in 5.3.1, finish A or finish B.

5.3.2.1 Accessories. Accessories, when provided, shall be packed as specified in 5.3.1.1, except that the box shall be domestic type, style optional. Strapping need not be zinc coated.

5.3.3 Commercial. The air conditioner and accessories, preserved as specified in 5.2.3, shall be packed in accordance with ASTM D 3951.

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5.4 Marking.

5.4.1 Military. Marking for military level of protection (level A or level B) shall be in accordance with MIL-STD-129, and the special marking specified in 5.4.3.

5.4.2 Commerical. Marking for commercial packaging shall be in accordance with ASTM D 3951. Weight and cube data shall be marked on each shipping container. The special marking specified in 5.4.3 shall be marked on all containers.

5.4.3 Special marking. For all degress of packing, special marking shall be placed on shipping containers to indicate that the air conditioner shall remain in an upright position during shipment and storage. Special marking shall consist of arrows and the words "THIS SIDE UP" which shall be placed on two sides of the container in black letters not less than one inch high.

6. NOTES

6.1 Intended use. The air conditioner is intended for use in providing cooled air for proper functioning of primary van-housed equipment and for comfort of operating personnel. The air conditioner may be installed indoors or may be exposed to the weather.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. Time frame required for submission of the first article model and number required (see 3.2).
d. When the Government will conduct any or all of the first article model examination and tests. When the Government will conduct some but not
all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and test shall be conducted by the contractor (see 3.2).
e. Color required (see 3.17).
f. When the accessories shall be furnished (see 3.18).
g. Time frame required for submission of the first article pack (see 5.1).
h. Degree of preservation and degree of packaging required (see 5.2 and 5.3).
i. When other than finish B strapping is required (see 5.3.1, 5.3.1.1, and 5.3.2).

6.3 First article models. When a first article inspection is required, the items should be initial production units. The first article should consist of two units. The contracting officer should include specific instructions in

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acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the document's first articles.

6.4 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the air conditioners in accordance with this specification.

6.5 Quality assurance provisons (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrective action taken when applicable.

6.6 Definitions.

6.6.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.6.2 End item final inspection requirement (EIFIR). An EIFIR is a check list of quality characteristics prepared by the Government which the contractor must follow during his inspection of the end item to verify complete and functional conformance to contract requirements prior to submission for final acceptance.

6.7 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts list, and contractor's maintenance and operational manual to be furnished with each air conditioner. In addition the contracting officer should require that copies of the performance curves specified in the run-in test be furnished to the engineering agency within 30 days after their development.

6.8 Finish paint coat validation. Prior to acceptance of any camouflage finish coat the contracting officer should require validation of spectral reflectance characteristics of the finish coat from the U.S. Army Belvoir Research and Development Center, ATTN: STRBE-VO, Fort Belvoir, VA 22060-5606.

6.9 Electromagnetic interference. This air conditioner has been previously procured and qualified to the requiements specified in section 3 and 4 of this specification. These are performance requirements only. Specific drawings which are incorporated therein are intended as an example of a method which was

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successful in the past. Since each contractor has latitude in selecting electrical components within the requirements of the drawings and specifications, and the electromagnetic emission and suppression characteristics may differ from those previously used, the Government does not expressly or impliedly warrant that performance requirements will be met if such drawings are followed and reliance upon such drawings is at the contractor's own risk. Bidders are cautioned against failing to properly take into consideration the radio frequency interfernece suppression/electromagnetic characteristics requirements for this air conditioner when preparing their bids.

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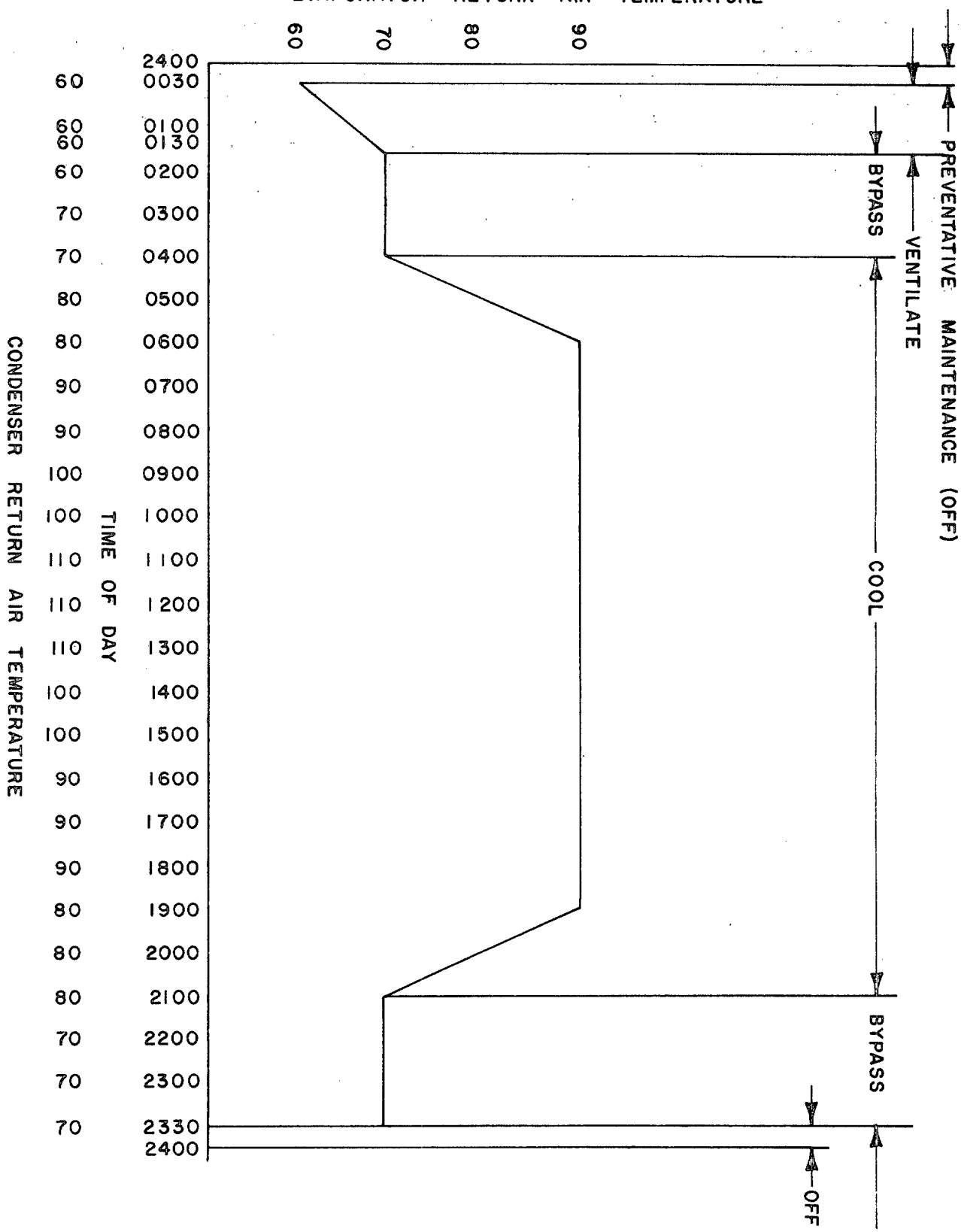


Figure 1. AIR CONDITIONER TEMPERATURE TIME TEST PROFILE

X-2332

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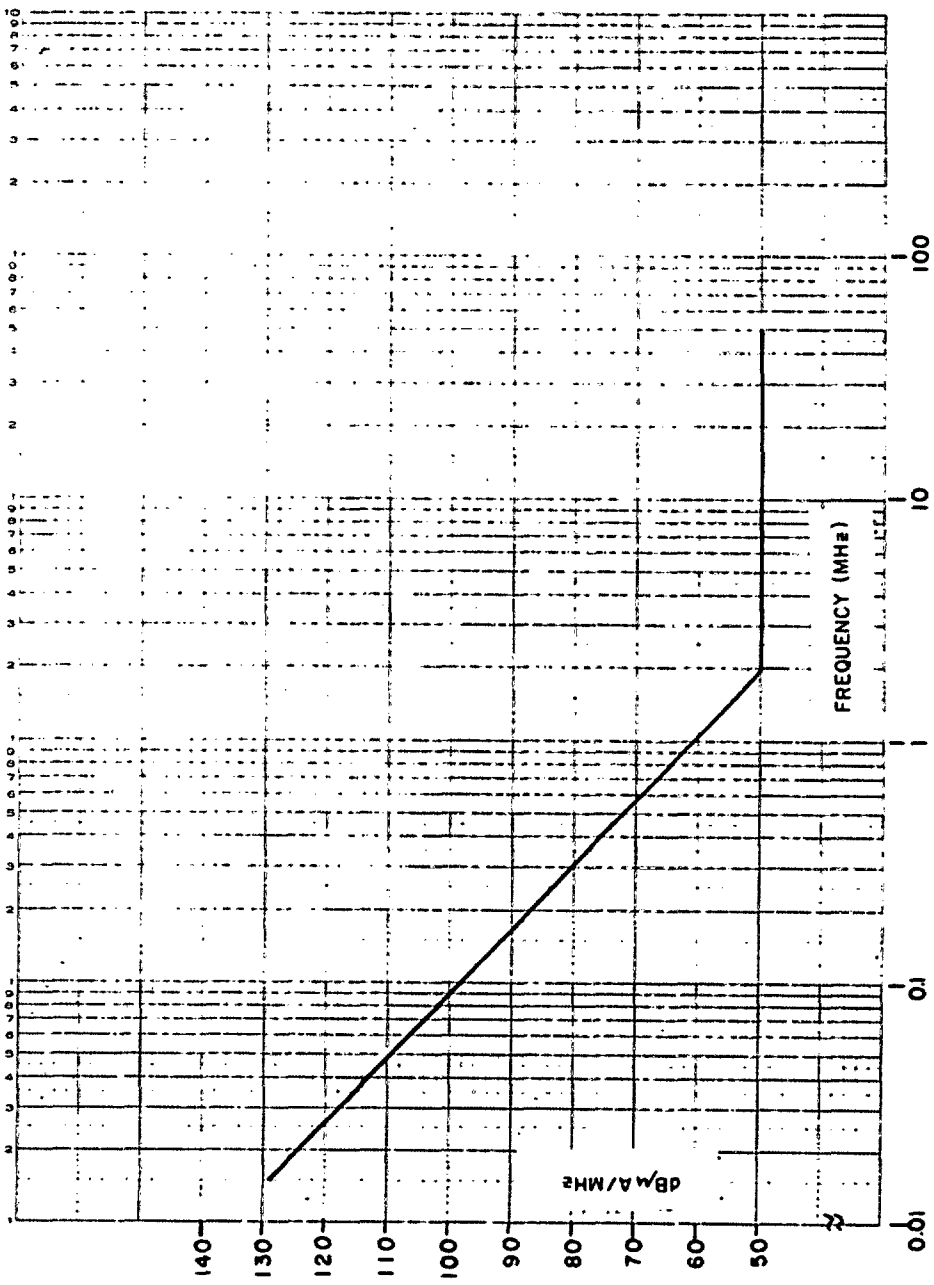


FIGURE 2. Limit for CE03 broadband emissions DC and interconnecting leads and AC leads.

X-3747

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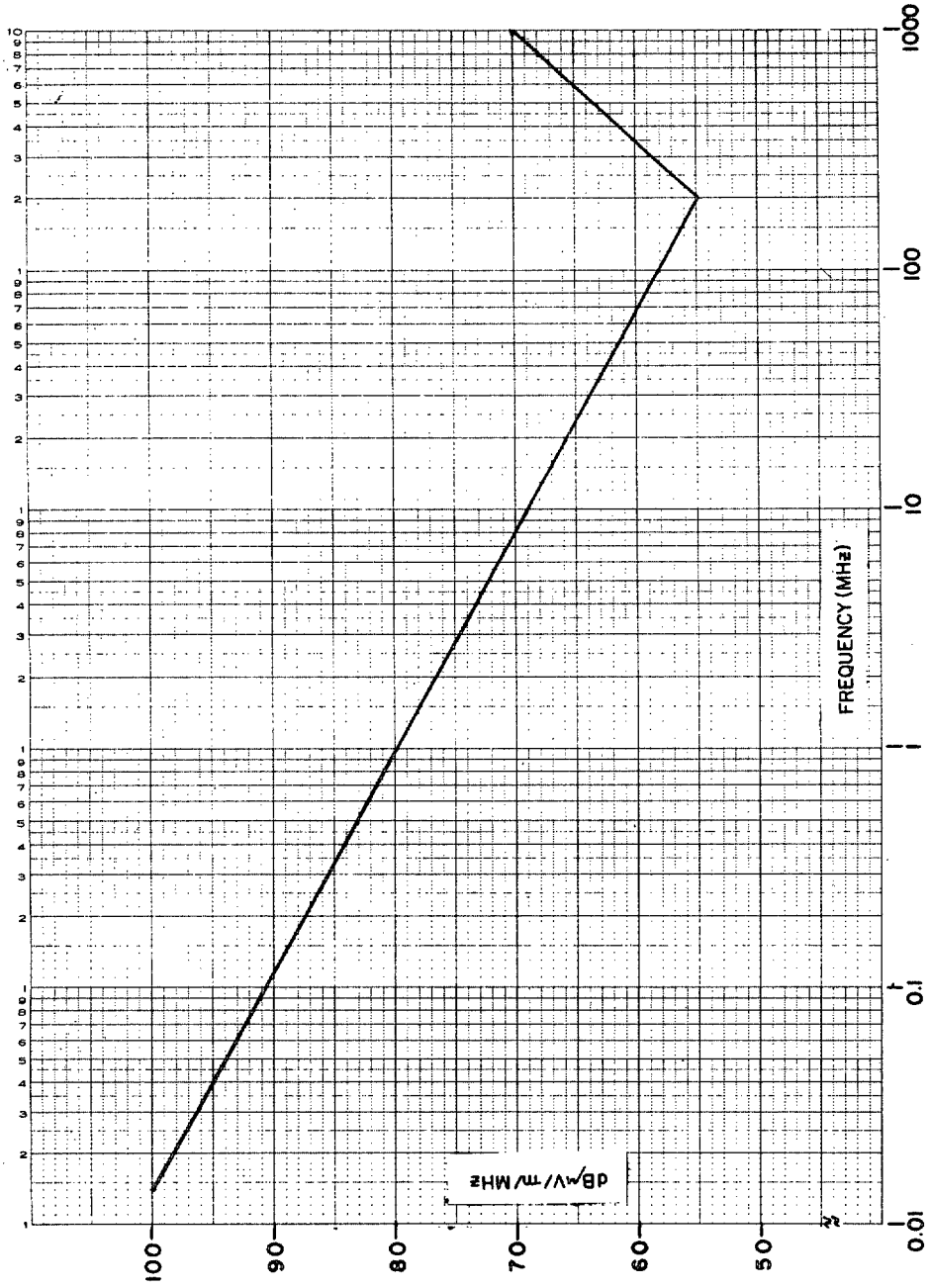
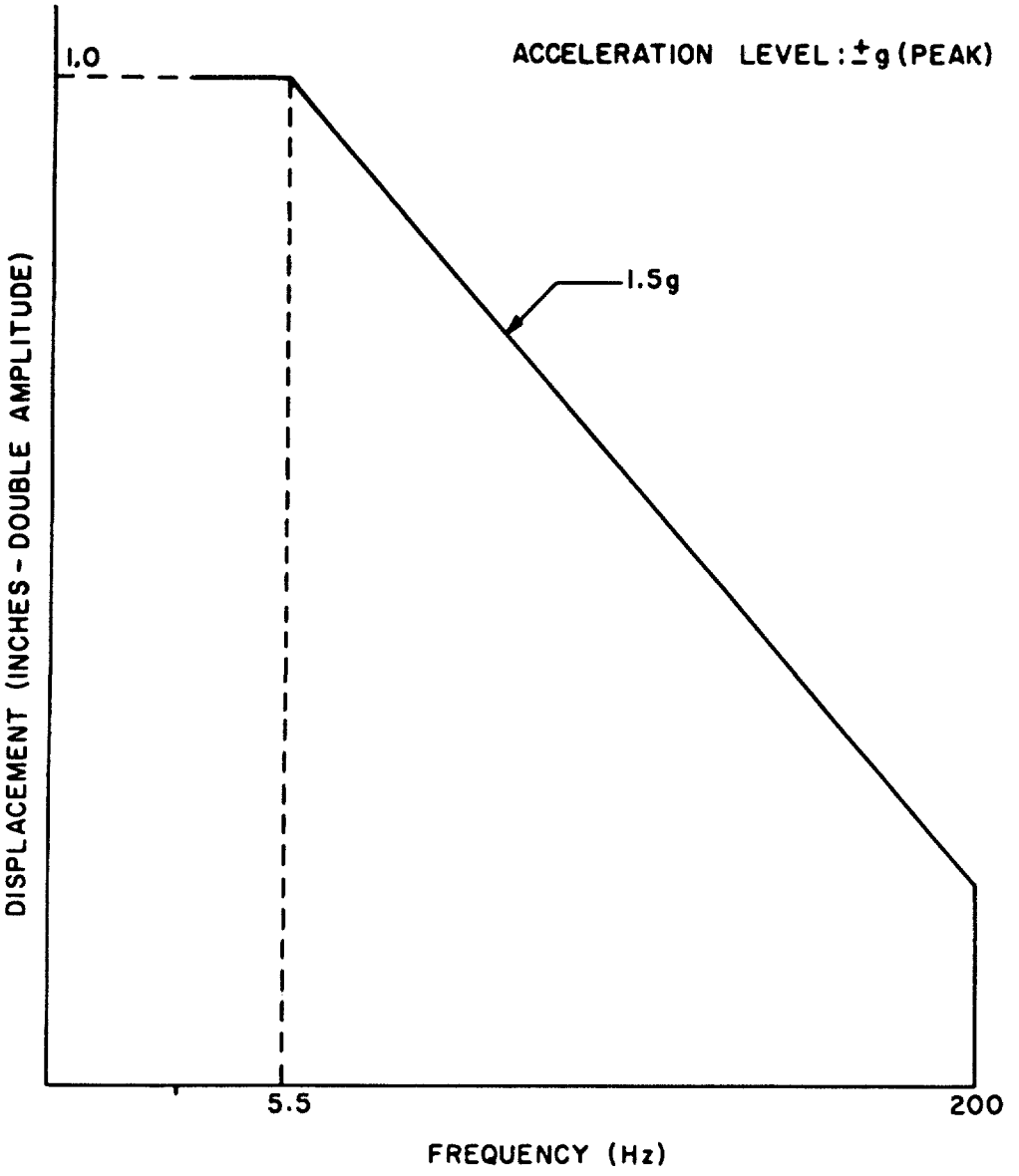


FIGURE 3. Limit for RE02 broadband emissions.

X-3748

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NOTE: THE CURVE SHALL BE EXTENDED TO 2 Hz WHEN TEST ITEM RESONANCES BELOW 5 Hz ARE EXPECTED.

FIGURE 4. Vibration test curve.

X-4276

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-52435D(ME)	Air Conditioner: Base Mounted, Air-Cooled, 208-Volt, 3-Phase, 50- and 60-Hertz, AC, Single-Package, 36,000BTU/Hr

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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